

EGH404 - Assessment 1:

Research Topic & Literature Critique

Generally, a Literature Critique is part of a process to conduct research on question(s) relating to a research problem. It identifies and assesses the content of literature on the topic and then narrows down and outlines how these findings relate to the research problem.

Literature critiques can focus on different parts of research – including research methods. When reviewing research methods, try to **focus on why the research methods selected by the authors were chosen**. Make sure to consider the followings:

- the objectives of the research (what specific research aspect(s) are the authors trying to address and solve),
- how the chosen method directly relates to the ways in which the research problem manifest itself and how the available data was collected and analysed; and
- suggestions for correction / improvement

Effective critiques will cover the article's significant and related points. A critique that ignores the article's main argument or concentrates on ideas mentioned parenthetically, suggests a lack of comprehension on the writer's part. Avoid composing a critique that solely addresses the article's strengths; such critiques tend to be read as summaries rather than assessments. Although you need not present a balanced opinion of the article, your critique must be fair.

You must recite the article's key ideas in your critique and quote sparingly; for example, only use quotes when the authors' language reveals distinctive phrasing or a lack of clarity. Summarizing or paraphrasing points allows you to demonstrate your understanding of the authors' arguments.

EGH404 Assessment 1 Literature Critique

Student name	Rory Blair
Student number	n10773398

Assessment 1 is worth 30 marks in total.

This is the template for Assessment 1 and there are 5 items to complete. Please ensure to follow and use this template when submitting your Assessment. Please also keep to the recommended word count (+/- 10%); please note that the total word count excludes references (including in-text references).

1 Introduce Your Research Topic (200 words total)

Please introduce your selected research topic and complete the 3 items below in your own words.

Please note that students can select a research topic from the list of projects posted on Canvas OR students can select their own research topic provided:

- (a) the topic is relevant to their engineering discipline; **AND**
- (b) the topic's research content and attributes have been identified in your research literature searches completed during the practical activities

1.1 Introduce your Research Topic. (< 150 words)

Use this section to provide background information on your research topic; you may use bullet point form.

- The research topic being investigated is Foreign Object Debris (FOD) Detection using Artificial Intelligence (AI)
- A FOD is an object that is in a location that may result in damage to a vehicle (aircraft) or a human.
- FOD incidents are rare but in some cases are fatal, like the Concorde Air France 4590 incident where a piece of debris from a different aircraft, slashed the tyre of the Concorde. The debris from the tyre then caused one of the fuel tanks to rupture, causing the incident ([Federal Aviation Administration, 2000](#))
- There are methods used to mitigate FOD incidents, such as physical checking by eye and sweepers, which include human interaction that could put the people performing these assessments in danger.
- There are AI's that have the capability to identify anomalies within images and videos.
- There are institutions that have developed designs for FOD detection using AI, however only one has an implemented design used at some airports. ([ArgosAI, 2024](#))

1.2 State your Research Problem. (<25 words)

In one sentence, state the problem your research project is aiming to solve.

- The problem is although there is an implemented design, it doesn't require verification from humans before a team is sent to the runway for inspection.

1.3 Identify your research question? (<25 words)

In one sentence, identify your project's research question.

- Research Question: Does AI need human verification when it comes to FOD detection on runways?
- This research question is aimed at the problem mentioned above

2 Demonstrate Your Literature Search

Please demonstrate your literature search methodology using 3 different **academic** database sources. **Please ensure that at least 1 database is specific to engineering.** Please note that non-academic databases (including ChatGPT are not permitted).

For each database you have selected to search, please copy & paste screenshot(s) of your database search page(s) that shows (a) search terms and filters you used when searching for your journal papers; and (b) top 10 results returned from your search.

Please ensure that your screenshots are clear and legible. You are permitted to copy/paste more than 1 image and use as much space as needed. Marks will be deducted if your screenshots are too difficult to read.

2.1 Database 1 Search: IEEE Xplore

Please copy & paste screenshot(s) of your database search page(s) that shows (a) search terms and filters you used when searching for your journal papers; and (b) top 10 results returned from your search.

A).

The screenshot displays the IEEE Xplore Advanced Search interface. The browser address bar shows the URL `ieeexplore.ieee.org/search/advanced`. The page title is "Advanced Search". There are three tabs: "Advanced Search" (selected), "Command Search", and "Citation Search".

Under the "Advanced Search" tab, the instruction "Enter keywords and select fields." is shown. The search criteria are as follows:

- Search Term: Foreign Object Debris, in: All Metadata
- AND
- Search Term: Detection, in: All Metadata
- AND
- Search Term: AI, in: All Metadata

Below the search criteria, the "Publication Date" section is visible with two options:

- ☐ Date Added to IEEE Xplore: From: mm/dd/yyyy To: mm/dd/yyyy
- ☐ Publication Year: From: 1884 To: 2024

At the bottom right, there are "Reset All" and "Search" buttons. On the right side, there is a "Preferences" section with links for "Learn More", "Data Fields", "Search Examples", "Search Operators", "Search Tips", and "Wildcard Limits". A promotional banner for "Boolean operators" is also present, along with an "IEEE" logo and a "Feedback" button.

B).

Showing 1-5 of 5 results for ("All Metadata":Foreign Object Debris) AND ("All Metadata":Detection) AND ("All Metadata":AI) ×

☐ Conferences (5)



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the IEEE Open Journal
of Control Systems

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☐ Open Access Only

Year

☒ Range

☐ Single Year

2021

2024

[Clear](#)

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☐ Select All on Page

Sort By [Relevance](#)

☐ [Live Detection of Foreign Object Debris on Runways Detection using Drones and AI](#)

Adam Parker; Felipe Gonzalez; Peter Trotter

2022 IEEE Aerospace Conference (AERO)

Year: 2022 | Conference Paper | Publisher: IEEE

Cited by: [Papers \(9\)](#)

[Abstract](#) [HTML](#) [PDF](#) [CC](#)

☐ [Performance Analysis of YOLOv4 and SSD Mobilenet V2 for Foreign Object](#)

☐ [Performance Analysis of YOLOv4 and SSD Mobilenet V2 for Foreign Object Debris \(FOD\) Detection at Airport Runway Using Custom Dataset](#)

Muhammad Reza Fairuzi; Fitri Yuli Zulkifli

2021 17th International Conference on Quality in Research (QIR): International Symposium on Electrical and Computer Engineering

Year: 2021 | Conference Paper | Publisher: IEEE

Cited by: [Papers \(3\)](#)

[Abstract](#) [HTML](#) [PDF](#) [CC](#)

☐ [UAV and AI Application for Runway Foreign Object Debris \(FOD\) Detection](#)

Ellena Papadopoulos; Felipe Gonzalez

2021 IEEE Aerospace Conference (50100)

Year: 2021 | Conference Paper | Publisher: IEEE

Cited by: [Papers \(14\)](#)

[Abstract](#) [HTML](#) [PDF](#) [CC](#)

☐ [Integrating AI and Computer Vision for Effective Foreign Object Detection on Airport Runways](#)

Annie Silviya S H; Kingsley S; Agitha W; Srirani B; Priyadarshan V J; Mohana V

2023 3rd International Conference on Pervasive Computing and Social Networking (ICPCSN)

Year: 2023 | Conference Paper | Publisher: IEEE

[Abstract](#) [HTML](#) [PDF](#) [CC](#)

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- ☐ **Integrating AI and Computer Vision for Effective Foreign Object Detection on Airport Runways** 
- Annie Silviya S H; Kingsley S; Agitha W; Sriman B; Priyadarshan V J; Mohana V
2023 3rd International Conference on Pervasive Computing and Social Networking (ICPCSN)
Year: 2023 | Conference Paper | Publisher: IEEE
- [✓ Abstract](#) [HTML](#)  
- ☐ **Autonomous Landing Guidance Assist for Aircrafts in Zero Visibility Conditions Using Model Based and AI Methods** 
- Mithun Sarkar; B. Anbarasu
2024 International Conference on Recent Advances in Electrical, Electronics, Ubiquitous Communication, and Computational Intelligence (RAEEUCCI)
Year: 2024 | Conference Paper | Publisher: IEEE
- [✓ Abstract](#) [HTML](#)  

2.2 Database 2 Search: Scopus

Please copy & paste screenshot(s) of your database search page(s) that shows (a) search terms and filters you used when searching for your journal papers; and (b) top 10 results returned from your search.

A).

Advanced query ☐

Search within

Article title, Abstract, Keywords

▼

Search documents *

foreign AND object AND debris

×

🗑

AND ▼

Search within

Article title, Abstract, Keywords

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Search documents

ai

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AND ▼

Search within

Article title, Abstract, Keywords

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Search documents

detection

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🗑

Save search

Set search alert

+ Add search field

Reset

Search 🔍

Document type

^

Conference paper

8

Documents

Beta

Preprints

Patents

Secondary documents

Research data ↗

Are you searching for: (TITLE-ABS-KEY (foreign AND object AND debris) AND TITLE-ABS-KEY (air) AND TITLE-ABS-KEY (detection))

^

8 documents found

Analyze results ↗

B).

Are you searching for: (TITLE-ABS-KEY (foreign AND object AND debris) AND TITLE-ABS-KEY (air) AND TITLE-ABS-KEY (detection))

^

8 documents found

Analyze results ↗

Refine search

Search within results

Filters

Year

Range

Individual

from

to

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Author name

Subject area

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All ▼

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Citation overview

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More

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Show all abstracts

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Sort by Date (newest) ▼

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Document title

Authors

Source

Year

Citations

☐

1

Conference Paper • Open access

Towards Small Anomaly Detection

Messerer, T.

International Conference on Pattern Recognition Applications and Methods, 1, pp. 860–865

2024

0

Show abstract ▼

Check for fulltext at QUT ↗

View at Publisher ↗

Related documents

☐

2

Conference Paper

Integrating AI and Computer Vision for Effective Foreign Object Detection on Airport Runways

Annie Silviya, S.H., Kingsley, S., Agitha, W., ... Priyadarshan, V.J., Mohana, V.

Proceedings - 2023 3rd International Conference on Pervasive Computing and Social Networking, ICPCSN 2023, pp. 382–387

2023

0

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Related documents

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3

Conference Paper • Open access

Live Detection of Foreign Object Debris on Runways Detection

Parker, A., Gonzalez, F.,

IEEE Aerospace

2022

9

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Check for fulltext at QUT ↗

View at Publisher ↗

Related documents

☐ Engineering	6	☐ 3	Conference Paper • Open access Live Detection of Foreign Object Debris on Runways Detection using Drones and AI	Parker, A. , Gonzalez, F. , Trotter, P.	IEEE Aerospace Conference Proceedings , 2022-March	2022	9
☐ Computer Science	4						
☐ Earth and Planetary Sciences	2						
☐ Materials Science	2						
☐ Mathematics	2	☐ 4	Conference Paper • Open access UAV and AI Application for Runway Foreign Object Debris (FOD) Detection	Papadopoulos, E. , Gonzalez, F.	IEEE Aerospace Conference Proceedings , 2021-March, 9438489	2021	15
Show all							
Document type	^						
☒ Conference paper	8						
Source title	v	☐ 5	Conference Paper Performance Analysis of YOLOv4 and SSD Mobilenet V2 for Foreign Object Debris (FOD) Detection at Airport Runway Using Custom Dataset	Fairuzi, M.R. , Zulkifli, F.Y.	17th International Conference on Quality in Research, QIR 2021: International Symposium on Electrical and Computer Engineering, pp. 11–16	2021	3
Publication stage	v						
Keyword	^						
☐ Object Detection	6						
☐ Foreign Object Debris	5						
☐ Debris	4						
☐ Aircraft Detection	3	☐ 6	Conference Paper A defect prevention concept using artificial intelligence	Eddy, D. , Krishnamurty, S. , Grosse, I. , White, M. , Blanchette, D.	Proceedings of the ASME Design Engineering Technical Conference, 9, V009T09A040	2020	3
☐ Deep learning	3						
Cancel							
Exclude							
Limit to							

☐ Foreign Object Debris	5								
☐ Debris	4								
☐ Aircraft Detection	3	☐	6	Conference Paper A defect prevention concept using artificial intelligence	Eddy, D., Krishnamurty, S., Grosse, I., White, M., Blanchette, D.	Proceedings of the ASME Design Engineering Technical Conference, 9, V009T09A040	2020		3
☐ Deep Learning	3								
Show all				Show abstract	Check for fulltext at QUT	View at Publisher	Related documents		
Affiliation									
Funding sponsor		☐	7	Conference Paper Recent advances in artificial intelligence applications to composites fabrication	Blake, S.	CAMX 2019 - Composites and Advanced Materials Expo, TP19-0658	2019		2
Country/territory				Show abstract	Check for fulltext at QUT	Related documents			
Source type									
Language		☐	8	Conference Paper Elements and mechanisms for applying artificial intelligence to composites fabrication	Blake, S.	International SAMPE Technical Conference, 2019-May	2019		7
☐ English	8			Show abstract	Check for fulltext at QUT	View at Publisher	Related documents		

Display 10 results [Back to top](#)

Export filter counts

2.3 Database 3 Search: Engineering Village (Compendex and Inspec)

Please copy & paste screenshot(s) of your database search page(s) that shows (a) search terms and filters you used when searching for your journal papers; and (b) top 10 results returned from your search.

A).

The screenshot displays the Engineering Village search interface. The search terms are: Foreign Object Debris, Detection, AI, and Runway. The filters include: Document type (Conference proceeding, Report chapter, Book, Dissertation, Report review, Book chapter, Journal article, Standard, Conference article, Preprint), Databases (All, Compendex, Inspec, GeoRef, US Patents, EP Patents, WO Patents, Knovel), and Sort by (Relevance). The search results show 10 records found in Compendex & Inspec for 1884-2024.

engineeringvillage.com/search/quick.url?SEARCHID=282aca2db3b6486eae15af8cc838302d&COUNT=1&usageOrigin=&usageZone=

Engineering Village

Search Search history 7 Alerts 0 Selected records 0 More 7 Create account

Quick search All fields for Foreign Object Debris

AND All fields for Detection

AND All fields for AI

AND All fields for Runway

Turn off AutoSuggest | + Add search field | Reset form

Databases Date Language Document type Sort by Browse indexes Autostemming Discipline Treatment

10 records found in Compendex & Inspec for 1884-2024: (((((Foreign Object Debris) WN ALL) AND ((Detection) WN ALL)) AND ((AI) WN ALL)) AND ((Runway) WN ALL)) AND (((CA) OR (CP) OR (JA) OR (RR)) WN DT)

Create alert Save search Share search RSS feed Sort by: Relevance

Databases Date Language Document type Sort by Browse indexes Autostemming Discipline Treatment

☐ All Document types ☐ Book ☐ Book chapter ☒ Conference article

☒ Conference proceeding ☐ Dissertation ☒ Journal article ☐ Preprint

☐ Report chapter ☒ Report review ☐ Standard

Databases Date Language Document type Sort by Browse indexes Autostemming Discipline Treatment

☐ All ☒ Compendex ☒ Inspec ☐ GeoRef ☐ US Patents ☐ EP Patents ☐ WO Patents ☐ Knovel

B).

engineeringvillage.com/search/quick.url?SEARCHID=282aca2db3b6486eae15af8cc838302d&COUNT=1&usageOrigin=&usageZone=

Engineering Village

Search Search history Alerts Selected records More Create account

10 records found in Compendex & Inspec for 1884-2024: (((((Foreign Object Debris) WN ALL) AND ((Detection) WN ALL)) AND ((AI) WN ALL)) AND ((Runway) WN ALL)) AND (((CA) OR (CP) OR (JA) OR (RR)) WN DT)

Create alert Save search Share search RSS feed

Sort by: Relevance

Display: 25 results per page

Refine

Remove duplicates

By physical property

Filter results by physical properties such as size, temperature, pressure and many more

By category

Download all

Limit to Exclude

Add a term

Open Access

All Open Access 3

Green 3

Document type

Conference article 10

1. Performance Analysis of YOLOv4 and SSD Mobilenet V2 for Foreign Object Debris (FOD) Detection at Airport Runway Using Custom Dataset

Fairuzi, Muhammad Reza (Department of Electrical Engineering Faculty of Engineering, Universitas Indonesia, Depok, Indonesia); Zulkifli, Fitri Yuli Source: 17th International Conference on Quality in Research, QIR 2021: International Symposium on Electrical and Computer Engineering, p 11-16, 2021, 17th International Conference on Quality in Research, QIR 2021: International Symposium on Electrical and Computer Engineering Database: Compendex Document type: Conference article (CA) Show preview Cited by in Scopus (3) Full text QUT Fulltext Finder

2. Live Detection of Foreign Object Debris on Runways Detection using Drones and AI (Open Access)

Parker, Adam (School of Electrical Engineering and Robotics, Queensland University of Technology (QUT), Brisbane; QLD; 4000, Australia); Gonzalez, Felipe; Trotter, Peter Source: IEEE Aerospace Conference Proceedings, v 2022-March, 2022, 2022 IEEE Aerospace Conference, AERO 2022 Database: Compendex Document type: Conference article (CA) Show preview Cited by in Scopus (9) Full text QUT Fulltext Finder

3. UAV and AI Application for Runway Foreign Object Debris (FOD) Detection (Open Access)

Feedback

engineeringvillage.com/search/quick.url?SEARCHID=282aca2db3b6486eae15af8cc838302d&COUNT=1&usageOrigin=&usageZone=

Engineering Village

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Document type

Conference article 10

Database

Compendex 5

Inspec 5

Author

Anbarasu, B. 2

Gonzalez, F. 2

Gonzalez, Felipe 2

Agitha, W. 1

Annie, Silviya S. H. 1

View more

Author affiliation

Controlled vocabulary

Object Detection 8

Computer Vision 5

Airports 4

Debris 4

Aircraft Detection 3

1. UAV and AI Application for Runway Foreign Object Debris (FOD) Detection (Open Access)

Papadopoulos, Ellena (School of Science and Engineering, Queensland University of Technology, Brisbane, Australia); Gonzalez, Felipe Source: IEEE Aerospace Conference Proceedings, v 2021-March, March 6, 2021, 2021 IEEE Aerospace Conference, AERO 2021 Database: Compendex Document type: Conference article (CA) Show preview Cited by in Scopus (15) Full text QUT Fulltext Finder

2. UAV and AI Application for Runway Foreign Object Debris (FOD) Detection

Papadopoulos, E. (Queensland University of Technology, School of Science and Engineering, Brisbane, QLD, Australia); Gonzalez, F. Source: 2021 IEEE Aerospace Conference, p 8 pp., 2021 Database: Inspec Document type: Conference article (CA) Show preview Cited by in Scopus (15) Full text QUT Fulltext Finder

3. Performance Analysis of YOLOv4 and SSD Mobilenet V2 for Foreign Object Debris (FOD) Detection at Airport Runway Using Custom Dataset

Fairuzi, M.R. (Universitas Indonesia, Department of Electrical Engineering, Indonesia); Zulkifli, F.Y. Source: 2021 17th International Conference on Quality in Research (QIR): International Symposium on Electrical and Computer Engineering, p 11-16, 2021 Database: Inspec Document type: Conference article (CA) Show preview Cited by in Scopus (3) Full text QUT Fulltext Finder

4. Live Detection of Foreign Object Debris on Runways Detection using Drones and AI (Open Access)

Parker, A. (Queensland University of Technology, School of Electrical Engineering and Robotics, Brisbane, QLD 4000, Australia); Gonzalez, Felipe; Trotter, Peter Source: IEEE Aerospace Conference Proceedings, v 2022-March, 2022, 2022 IEEE Aerospace Conference, AERO 2022 Database: Compendex Document type: Conference article (CA) Show preview Cited by in Scopus (9) Full text QUT Fulltext Finder

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Engineering Village

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More

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A

☐ Debris

4

☐ Aircraft Detection

3

View more

Classification code

Country/Region

Language

Year

Source title

Publisher

Limit to

Exclude

New search with facets

6. ☐ Live Detection of Foreign Object Debris on Runways Detection using Drones and AI (Open Access)

Parker, A. (Queensland University of Technology, School of Electrical Engineering and Robotics, Brisbane, QLD 4000, Australia); Gonzalez, F.; Trotter, P. Source: 2022 IEEE Aerospace Conference (AERO), p 13 pp., 2022

Database: Inspec

Document type: Conference article (CA)

Show preview Cited by in Scopus (9) Full text QUT Fulltext Finder

7. ☐ Integrating AI and Computer Vision for Effective Foreign Object Detection on Airport Runways

Annie Silviya, S.H. (Rajalakshmi Institute of Technology, Department of Computer Science and Engineering, Chennai, India); Kingsley, S.; Agitha, W.; Sriman, B.; Priyadarshan, V.J.; Mohana, V. Source: Proceedings - 2023 3rd International Conference on Pervasive Computing and Social Networking, ICPCSN 2023, p 382-387, 2023, Proceedings - 2023 3rd International Conference on Pervasive Computing and Social Networking, ICPCSN 2023

Database: Compendex

Document type: Conference article (CA)

Show preview Full text QUT Fulltext Finder

8. ☐ Integrating AI and Computer Vision for Effective Foreign Object Detection on Airport Runways

H, A.S.S. (Rajalakshmi Institute of Technology, Department of Computer Science and Engineering, India); S, K.; W, A.; B, S.; J, P.V.; V, M. Source: 2023 3rd International Conference on Pervasive Computing and Social Networking (ICPCSN), p 382-7, 2023

Database: Inspec

Document type: Conference article (CA)

Show preview Full text QUT Fulltext Finder

9. ☐ Autonomous Landing Guidance Assist for Aircrafts in Zero Visibility Conditions Using Model Based and AI Methods

Sarkar, M. (Dept of Aeronautical Engineering, Hindustan Institute of Technology and Science Padur, India); Anbarasu, B. Source: Conference on Recent Advances in Electrical, Electronics, Ubiquitous Communication, and Computational Intelligence (RAEEUCCI)

Feedback

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Search

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Alerts

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9. ☐ Autonomous Landing Guidance Assist for Aircrafts in Zero Visibility Conditions Using Model Based and AI Methods

Sarkar, M. (Dept of Aeronautical Engineering, Hindustan Institute of Technology and Science Padur, India); Anbarasu, B. Source: 2024 International Conference on Recent Advances in Electrical, Electronics, Ubiquitous Communication, and Computational Intelligence (RAEEUCCI), p 1-5, 2024

Database: Inspec

Document type: Conference article (CA)

Show preview Full text QUT Fulltext Finder

10. ☐ Autonomous Landing Guidance Assist for Aircrafts in Zero Visibility Conditions Using Model Based and AI Methods

Sarkar, Mithun (Hindustan Institute of Technology and Science Padur, Dept of Aeronautical Engineering, Chennai, India); Anbarasu, B. Source: 2024 International Conference on Recent Advances in Electrical, Electronics, Ubiquitous Communication, and Computational Intelligence, RAEEUCCI 2024, 2024 International Conference on Recent Advances in Electrical, Electronics, Ubiquitous Communication, and Computational Intelligence, RAEEUCCI 2024

Database: Compendex

Document type: Conference article (CA)

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1 of 1 pages

Display: 25 results per page

Engineering Village

Access

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3 Critique Empirical Research Article 1 (1300 words total)

Instructions: Select an empirical research article to critique and attach a PDF copy of the article when submitting your assessment. The article must be from peer reviewed journals and/or conference papers. Please note that reviews (including meta-analyses, surveys, scoping reviews, etc.) or book chapters are not permitted; if you are unsure, please ask a sessional staff member.

Please answer all items below. To aid with marking, please try to use bullet point form when answering this item*

Title of paper	UAV and AI Application for Runway Foreign Object Debris (FOD) Detection
APA (Conference Paper) reference	Papadopoulos, E., & Gonzalez, F. (2021). UAV and AI Application for Runway Foreign Object Debris (FOD) Detection. <i>2021 IEEE Aerospace Conference (50100)</i> (pp. 1-8). Big Sky, Montana, USA: IEEE.

3.1 Introduction (<200 words total):

- **Main Objectives set out in the Conference Paper:**
 - The main objective of this conference paper is to show the concept, testing, and results of a FOD detection system using AI and Unmanned Aerial Vehicles (UAV).
 - This concept involves using a UAV/Drone to fly the entire length of the runway, whilst recording the runway. This recording would then be studied, using AI, to make sure the runway is void of any FODs. This AI would need to differentiate between different materials, such as metal (Bolts and strips), and plastic (bottles and sheets) and still work in different types of weather.
 - This concept is tested using multiple different variables such as different types of drones or different types of AI and how that AI was been trained to detect FODs.
 - This concept is designed to have a high rate of accurate FOD detections, which would lead to fewer disruptions and smaller intervals of time where the runways are closed. (Papadopoulos & Gonzalez, 2021)
- **Research Question being Addressed by the Author(s):**
 - Although the research question is not stated, reading from the abstract shows that a possible research question is "Are UAVs, with AI assistance, a better method for FOD detection than current detection methods?"

3.2 Assessment (< 500 words in total):

- **Main Outcomes of the Research:**
 - Two AI models, 'Yolov3' and 'Microsoft Azure Custom Vision toolbox', were trained and tested in image/video analysis and anomaly recognition.
 - One outcome of the research was that the 'Microsoft Azure Custom Vision' AI was considered to be more reliable than 'Yolov3' when it came to FOD detection.
 - Another outcome of the research was that the items, whether clustered together or spread out, were for the majority of the test situations, correctly identified to their respective classes as noted below.
 - Clustered Accuracy:
 - Paper, 92%
 - Bottle, 78%
 - Plastic, 76%
 - Metal Strips, 72-83%
 - Metal Bolts, 85-97%
 - Spread out accuracy:
 - Paper, 98%.
 - Bottle, 89%
 - Plastic, 76%
 - Metal Strips, 81%
 - Metal Bolts, 82-94%
 - Plastic may be harder to identify as external factors, such as sunlight, may affect the camera's ability to detect plastic ([Papadopoulos & Gonzalez, 2021](#)).
- **Research Methods:**
 - The first method was the use of a 'Do It Yourself' (DIY) data set for training the AI, which allows for a variety of different pictures to be analysed. This would include different object types, object sizes, number of objects, and where they were placed.
 - The second method was to test which AI model would be used for the 'Full-Scale Testing'. This was completed in the 'Base Experimental Results' phase where a car park was used instead of a runway and varying flight levels were tested.
 - The final method was to test the optimum combination of drone and AI type, to test a real-world scenario.
- **Evidence for the Work Conducted:**
 - There is a diagram of how the concept would work and two diagrams of the airborne and ground elements involved in the drone.
 - The first table compares the different drone models. (each having varying capabilities and specifications) The second table is a test of the accuracy of the 'Yolov3' AI Model when the UAV is at different heights. This shows that the model is only good if the UAV is close to the ground. The third table is also a test of the accuracy of the 'Microsoft Azure Custom Vision' AI Model when the UAV is at different heights. This shows that the model is more accurate at higher flight levels than the 'Yolov3'.
 - The testing above, involves multiple screenshots being taken of the testing site, the AI model's working, and how they identify/labels objects.
- **Assumptions made:**
 - One assumption made by the author was that dependent on differing cloud formations and sunlight conditions, the results may be impacted.
 - In addition, the assumption was made that a height of 12m was best used for the final test on the runway, as this would provide the most accurate result. ([Papadopoulos & Gonzalez, 2021](#))
- **How the work Enhances Understanding of the research problem being investigated:**
 - This conference paper shows that even trained AI models are insufficient when it comes to FOD detection on runways.

3.3 Concluding commentary (< 300 words in total):

- **Strength of Conference Paper:**
 - One strength of the conference paper is that it show both AI models working and what the AI models identifies objects as.
 - Another strength is that the data set used to train both AI models was extensive, ensuring the AI models were well-trained before testing commenced.
 - Another strength is that the tables included within the paper contain all the necessary information so that the reader understands why certain decisions were made.
 - In addition, the video at the end of the paper shows how they created different scenarios and how the 'Microsoft Azure Custom Vision' model identifies the FOD in the images.
- **Weaknesses of Conference Paper:**
 - The height from the ground for the 'Full-Scale Testing' was unclear or not stated.
 - Although stated in the 'abstract' that clouds and sunlight could affect the results, there was no testing of the concept during more intense conditions, such as rain, wind, fog, or storms.
 - Another weakness of the paper was that a lot of the figure labelling is illegible and therefore difficult to decipher
 - A weakness for this paper is that the video attached did not display the 'Yolov3' AI model being experimented.
 - Another possible weakness is that only stationary object were used in testing whereas moving object would also impact test results.
- **Describe how it can inform/contribute to your research topic:**
 - This conference paper shows that even some of the AI models trained in FOD detection do need verification before a team is sent out to clear a runway.
 - Although not perfect, in certain circumstances, the concept does work with an accuracy of 100% when at an altitude of approximately 3m (as opposed to the 12m assumed in the paper).

3.4 Annotated Bibliography (< 300 words in total):

Identify two (2) useful sources from the reference list of your paper and briefly summarise how the selected sources can inform/contribute/address your research topic.

Reference 1 Citation (APA Report):	INSIGHT SRI Ltd. (2008). <i>The Economic Cost of FOD to Airlines</i> . London, UK: INSIGHT SRI Ltd.
Reference 1 Summary (<150 words):	• This reference is a report that displays the cost airlines and airports have waisted due to FOD incidents. • This report goes into very particular detail by showing what type of damage was done to an aircraft and what this damage was caused by. • Since tyres/engines usually come with a warranty for FOD, not only are airlines and airports losing money, but the manufacturers of said parts are also losing money (INSIGHT SRI Ltd, 2008). • The report also considers FODs to be safety hazards and nuisances. • In the summary of this report, they state that FOD causes airlines per airport per year a cost of around USD 20 Million, which is usually considered the "cost of doing business" for most airlines. (INSIGHT SRI Ltd, 2008) • Considering this report was from 2008, it is safe to say that the 'cost of doing business' has increased. • This report is evidence to support that the threat of FODs on runways must be eliminated.

Reference 2 Citation (APA Conference Paper):	Ang, L. (2013). Research and Design of an Airfield Runway FOD Detection System Based on WSN. <i>International Journal of Distributed Sensor Networks</i> , 1-12.
Reference 2 Summary (<150 words):	• This reference is another conference paper that conveys a different concept for FOD detection. • This concept would use a Wireless Sensor Network (WSN), which would include the use of cameras, millimeter wave radar, laser range finders, photoelectric encoder, and a data fusion device. (Ang, 2013) • Workflow of the proposed system: 1). Setup of Hardware, Software and team 2). Radar begins operation checking runway 3). Once FOD is detected, other sensors would be utilised to verify radar data. 4). If other sensors are unable to correlate with radar, manual inspection is recommended. 5). Record incidents in the database. • This paper provides more evidence to support the importance of research for more secure and accurate FOD detection methods. • As stated in the 'conclusion and future work' of this paper, research should be placed in 'real-time' detection of FODs as well as better image analysis algorithms.

4 Critique Empirical Research Article 2 (1300 words total)

Instructions: Select an empirical research article to critique and attach a PDF copy of the article when submitting your assessment. The article must be from peer reviewed journals and/or conference papers. Please note that reviews including meta-analyses, surveys, scoping reviews, etc. or book chapters are not permitted; if you are unsure, please ask a sessional staff member.

Please answer all items below. To aid with marking, please try to use bullet point form when answering this item*

Title of paper	Live Detection of Foreign Object Debris on Runways Detection using Drones and AI
APA (Conference Paper) reference	Parker, A., Gonzalez, F., & Trotter, P. (2022). Live Detection of Foreign Object Debris on Runways Detection using Drones and AI. 2022 <i>IEEE Aerospace Conference (AERO)</i> (pp. 1-13). Big Sky, Montana, USA: IEEE.

4.1 Introduction (<200 words total):

- **Main Objectives set out in the Conference Paper:**
 - The main objective is to explain the concept being proposed, physically test the concept, and record the results. This concept is for a Foreign Object Debris (FOD) detection system using Unmanned Aerial Vehicles (UAV/drones) for a whole sweep of the runway and Artificial Intelligence (AI) for anomaly detection
 - Another objective of this conference paper is to focus on the detection of manmade and natural objects such as metal (manmade) and rock(natural). This paper does not talk about wildlife FODs such as animals.
 - This concept works in a similar way to the first empirical research article that was critiqued. However, they are very different in their approach to testing.
 - The design concept is to have a system that can correctly detect FOD objects on taxiways and runways to minimise money spent due to FOD incidents.
- **Research Question being Addressed by the Author(s):**
 - Although a research question is not immediately obvious, after reading the introduction and 'abstract' from the paper, the following research question is relevant to the research topic.
 - Research Question: "Are there alternative methods, utilising UAVs and AI, that would be more accurate and precise for FOD detection?"

4.2 Assessment (< 500 words in total):

- **Main Outcomes of the Research:**
 - During the training process, using the multiple detection models presented, it was discovered that the 'compact' AI models had an accuracy of 40-55%, whilst the 'general' AI models had an accuracy of 76-94% (Parker, Gonzalez, & Trotter, 2022).
 - Unlike the first empirical article, this paper only conducted the test with one AI model, 'Microsoft Azure Custom Vision'. This AI model used two different object, identification methods:
 - 1). The 'multi-general' model uses 6 different classes to identify certain types of objects on the runway. These six classes are called litter, wood, rubber, metal, bolt and nut. It was discovered that this model has an error rate of 32.1%. (Parker, Gonzalez, & Trotter, 2022)
 - 2). The 'single-general' model uses one class to identify the many classes stated in the 1st model. It was stated that the model had an error rate of 10.5%, which is better than the '6-class' model. (Parker, Gonzalez, & Trotter, 2022)
 - There are other types of AI models called 'multi-compact' and 'single-compact'. These AI models are faster at processing however seemed to have a higher error rate of 80%.
 - Another outcome was that the altitude of the drone did not change the effect of how it labelled objects on the image, but the altitude did change the amount of space the UAV would scan.
- **Research Methods:**
 - Three test sites were used for different parts of the experiment allowing for different runway environments to be tested:
 - 1). An abandoned airstrip in Noosa, which would allow for an environment where the runway may not be in the best condition.
 - 2). A Drone approved park, which may allow for an environment where the runway is maintained and kept in a superior condition.
 - 3). A 'drag strip' scenario that makes allowance for when the runway is reflecting sunlight due to either design or wet weather.
 - The UAVs were also tested to find which would be able to take the minimum number of images and still record the whole runway. This was done theoretically and practically.
- **Evidence for the Work Conducted:**
 - There is substantial evidence to show that physical experiments were conducted to help promote this concept of FOD detection.
 - Two UAVs were tested against each other.
 - Different altitudes were tested.
 - Graphs of how many objects were detected by the 'single/multi-general/compact' models.
 - The first graph shows the comparison of true positives of object detection using the 'multi-general' model
 - The second graph shows the comparison of true positives of object detection using the 'single-general' model.
- **Assumptions made:**
 - One assumption made by the author(s) during testing was that the UAV's ground speed would be around 25km/h, without wind affecting the flight.
 - Another assumption made by the authors was that a minimum of 1 pixel per centimeter is reasonable.
- **How the work Enhances Understanding of the research problem being investigated:**
 - This would increase the understanding of the research problem as it shows many ways FOD detection could be done.

4.3 Concluding commentary (< 300 words in total):

- **Strength of Conference Paper:**
 - One strength of this conference paper is that there were multiple testing sites (runway, park, drag strip), which allows testing for different types of runways and the environments around them.
 - Another strength for this paper is that it includes a link to a YouTube video. This video shows the concept working, as well as showing the four different types of AI model tested
 - In addition, the paper explains a particular situation where the runway material and weather combined, affecting the camera's ability to detect FODs.
- **Weaknesses of Conference Paper:**
 - The experiments carried out in the paper show that there is room for improvement in the detection process, and in conclusion, state that future works can be done to enhance FOD identification ([Parker, Gonzalez, & Trotter, 2022](#)). This includes:
 - Data augmentation in training (whilst training was sufficient, it can be upgraded to deliver more accurate results.)
 - Include other weather conditions for testing (cloud/rain/wind)
 - Increase object complexity and AI model (unknown objects)
 - Multi-spectral camera for more ways to detect FOD
 - In this experiment, no 'False Positive', 'False Negatives', 'True Positives' and 'True Negatives' have been considered. To improve the AI model and experiment outcome, these factors should be considered in the overall analysis.
 - While the conference paper states there were five pictures taken during the experiment, only two images were included as screenshots in the paper. No other images are included in an appendix
- **Describe how it can inform/contribute to your research topic:**
 - This paper can contribute to the research topic as it suggests alternative FOD detection methods that can be used in the future.
 - In summary, there appears there are several areas of this experiment that can be improved, to provide more time efficient and safer detection of FODs.

4.4 Annotated Bibliographies (< 300 words in total):

Identify two (2) useful sources from the reference list of your paper and briefly summarise how the selected sources can inform/contribute/address your research topic.

Reference 1 Citation (APA Journal Article):	Radha, M., & Hayder, H. (2021). Object Detection Under Rainy Conditions for Autonomous Vehicles: A Review of State-of-the-Art and Emerging Techniques. <i>IEEE Signal Processing Magazine - Computer Vision and</i> , 53-67.
Reference 1 Summary (<150 words):	• This reference is a report article that explores the way an object is detected by AI through rain, light or heavy. • The main objective was to present a seminar on emerging techniques to mitigate the effects of natural visual distortions for autonomous vehicles cameras to detect objects. (Radha & Hayder, 2021) • To complete this objective would include experimenting and analysing said results. This would also include an evaluation of an existing 'deraining' algorithm, and the algorithms strengths, weaknesses and limitation. The article also states that there is a shortage of 'deraining' techniques. • Results from many experimental methods tested are also included within. • The results show that if there are raindrops that are present on the lens when the analysis is complete, the algorithm does not identify any objects that are clearly seen. • This article would be useful as the problem that is being investigated in this article, could affect the ability for FOD detection.
Reference 2 Citation (APA Conference Paper):	Narasimhan, S. G., & Nayar, S. (2001). Removing weather effects from monochrome images. <i>Proceedings of the 2001 IEEE Computer Society Conference on Computer Vision and Pattern Recognition. CVPR 2001</i> (pp. 1-8). Kauai, Hawaii, USA: IEEE.
Reference 2 Summary (<150 words):	• This reference is a report that displays the technique used to eliminate bad weather condition which affects camera resolution and contrast, which would be useful for anomaly detection within images. • This technique does not need scene information or weather-specific image, and is effective with a variety of different weather conditions. This may include fog, haze, mist, smoke and other visually imparting weather conditions. (Narasimhan & Nayar, 2001) • This method could also be applied to other types of images, such as gray-scale, Infrared image, multi-spectral and RGB images. This method could also work with videos as well. • To complete these objectives theoretical models were calculated and physical experiments were also conducted. • The papers results show that the objective of this paper were met and the evidence to support this, is displayed within. • This paper would be essential as the problem this paper is trying to solve is one of many problems within FOD detection methods.

5 References (APA)

- Ang, L. (2013). Research and Design of an Airfield Runway FOD Detection System Based on WSN. *International Journal of Distributed Sensor Networks*, 1-12.
- ArgosAI. (2024). AFOD. Retrieved from ArgosAI: <https://argosai.com/foreign-object-detection/>
- Federal Aviation Administration. (2000). *Concorde, Air France Flight 4590, F-BTSC*. Paris, France: Federal Aviation Administration.
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- Radha, M., & Hayder, H. (2021). Object Detection Under Rainy Conditions for Autonomous Vehicles: A Review of State-of-the-Art and Emerging Techniques. *IEEE Signal Processing Magazine - Computer Vision and* , 53-67.